

Dietary Fat, Eicosanoids, and Immunity

PATRICIA V. JOHNSTON

*Department of Food Science and Division of Nutritional Sciences
University of Illinois at Urbana-Champaign
Urbana, Illinois*

I.	Introduction	103
II.	The Immune System and the Eicosanoids	105
	A. The Immune Response	105
	B. Membrane Structural Changes and Biochemical Events during Lymphocyte Activation	109
	C. Role of Phagocytes	114
	D. Eicosanoid Modulation of Immune Responses	117
III.	Dietary Fat and the Immune Responses	121
	A. Quality and Quantity of Fat	121
	B. Dietary $n - 6$ and $n - 3$ Fatty Acids and Eicosanoid Synthesis by Immunocompetent Cells	130
IV.	Conclusions	136
	References	139

I. Introduction

The evidence for the dietary essentiality of linoleic acid ($18:2n - 6$) or the $n - 6$ family of fatty acids is unequivocal. The absence of $18:2n - 6$ or its desaturated, elongated product in the diet is characterized by a classical deficiency syndrome which includes a scaly dermatitis, excessive water loss from the skin, impaired growth and reproduction, and poor wound healing. Numerous biochemical abnormalities are associated with the deficiency, including a decrease in $n - 6$ fatty acids in tissues and, if α -linolenic acid is also absent, an elevated $(20:3n - 9)/(20:4n - 6)$ ratio (Guarnieri and Johnson, 1970; Aaes-Jørgensen, 1982). While the α -linolenic acid ($18:3n - 3$) or $n - 3$ family is known to be essential for growth and development in various species of fish and shellfish, convincing evidence that these fatty acids have an essential function in mammals is lacking. Only one case of an apparent $18:3n - 3$ deficiency in man has been reported (Holman *et al.*, 1982), and extensive dietary studies using rats failed to show any effects on growth, reproduction, and general health (Tinoco *et al.*, 1978). Some evidence for neurological damage due to an $18:3n - 3$ deficiency does exist (Lamprey and Walker, 1976). For

many years, increased consumption of vegetable oils has been advocated for reduced risk of coronary heart disease. Consumption of vegetable oils by the Western world has increased dramatically since the early part of the century. As a consequence, the daily intake of much of the population may be 8 to 10 times higher than the 1–4 en% usually recommended. Little is known regarding the possible impact of this high intake of the dietary precursor of the 2-series eicosanoids on physiological functions. While eicosanoids are not stored and are produced only in response to stimuli, it is known that higher intakes of 18:2 $n-6$ lead to greater synthesis of prostaglandins in stimulated tissues (Mathias and Dupont, 1979). Until recently, scant attention was paid to the implications of changing the dietary ratio of $n-3$ fatty acids, the precursors of 3-series eicosanoids, to the $n-6$ fatty acids. In populations where soybean oil is the major vegetable oil, $n-3$ fatty acid intake has also increased, but far less than have the $n-6$ fatty acids. Many common vegetable oils contain less than 1% 18:3 $n-3$, and much of the soybean oil consumed has its 18:3 $n-3$ content reduced to 3–4% to improve stability. Epidemiological studies which showed that populations consuming a large amount of marine foods have less risk of coronary heart disease, prolonged bleeding times, and immune-related inflammatory disease (Dyerberg *et al.*, 1975; Dyerberg and Bang, 1979) led to a renewed interest in the $n-3$ fatty acids and research in the area accelerated rapidly.

Changes in the patterns of dietary polyunsaturated fatty acids can be expected to modify physiological functions by changes in cell membrane structure, in fluidity, and signal transmission. These effects with respect to lymphocyte function have been reviewed recently by Traill and Wick (1984). Gurr (1983) extensively reviewed the evidence for the influence of lipids on immune function. He included a brief section on the influence of prostaglandins and other oxygenated fatty acids, but dietary modulation of eicosanoid synthesis and immune function was not a major emphasis of his review. The focus of this article is the influence of dietary fat, particularly the essential fatty acids, on immune functions and how such influence may or may not be mediated via modified eicosanoid synthesis. Before the dietary studies are reviewed, brief accounts of the nature of the immune response and the immunoregulatory role of eicosanoids are given. An extensive coverage of these is beyond the scope of this review. More extensive discussion of the immune response, especially as it relates to nutrition, has been given by Chandra and Newberne (1977) and Stinnett (1983). An attempt to define the role of eicosanoids in the immune response is a highly complex task. Views differ and the present situation is confusing, although there is general agreement that eicosanoids have a role in lymphocyte and macrophage function. Three somewhat different

concepts of this role have been given by Goldyne and Stobo (1981), Goodwin and Webb (1980), and Parker (1981).

II. The Immune System and the Eicosanoids

A. THE IMMUNE RESPONSE

Vertebrates are protected from infections, mutant cells, and other foreign invaders, collectively known as antigens, by a complex system known as the "host defenses." These defenses may be immunological, involving cells of the immune system, or nonimmunological, including mucus, cilia, skin, the complement system of the blood, and interferon. In this article we will be concerned with immunological defense only.

The immune system is a system of organs, the circulatory lymphatics, the cells comprising these, and the multitude of molecules involved in the immune response. The response has a dual nature, although the two branches of the response are intimately intertwined. The responses are due to two major subclasses of lymphocytes, the T and the B cells, which both arise from the pluripotential cells of the bone marrow. Both have the ability to acquire a memory of encounters with specific antigens. The T (thymus-dependent) cells mature in the thymus and then migrate to various lymphoid organs. They do not have effector cell capability for fighting foreign invaders until they have been activated by antigenic stimuli such as viruses, parasites, tumor cells, or fungi. They are usually exposed to antigens by important accessory cells, the phagocytic macrophages. Once they are activated, metabolic changes occur culminating in cell division or transformation, with the formation of clones which share the memory of a specific antigen. Upon reexposure to the same antigen, they exert their effect by producing soluble lymphokines which have a wide range of activities which, with the aid of other cells, can lead to the disposal of the antigen. A partial list of lymphokines is given in Table I. This branch of the immune response is termed cell-mediated immunity or CMI. It is considerably more complex than described here in that T cells are subdivided into several classes which show specific functions, such as helper (T_H), suppressor (T_S), cytotoxic (T_C), and T_D cells responsible for delayed hypersensitivity in which some 24–48 hours after exposure of skin to antigen secondary responses start to occur. Delayed hypersensitivity is a useful, simple tool for testing for change in CMI. In the human it is a useful procedure to check immunocompetence. It involves detection of cutaneous hypersensitivity to an antigen or group of antigens. In the human a battery of recall antigens such as streptokinase–streptodornase,

Table I
SOME LYMPHOKINES AND THEIR ACTIVITIES

Lymphokine	Activity
Migration inhibitory factor	Inhibits macrophage movement <i>in vitro</i>
Macrophage aggregation factor	Agglutinates macrophages <i>in vitro</i>
Macrophage chemotactic factor	Attracts macrophages <i>in vitro</i>
Neutrophil, basophil, eosinophil chemo- tactic factors	Attract the cells <i>in vitro</i>
Leukocyte inhibitory factor	Inhibits cell movement <i>in vitro</i>
Lymphocyte blastogenic (mitogenic) factor	Induces lymphocyte DNA synthesis <i>in vitro</i>
Antibody production factors	Affect antibody production
Transfer factor	Transfers delayed hypersensitivity to specific antigens from one individual to another
Interferon	Prevents viral replication in target cells
Lymphotoxin	Destroys lymphocytes

Candida extract, and mumps is injected intradermally, the area is marked out, and 48 hours later the erythema and induration are measured. The test can be applied to laboratory animals by immunizing them against a foreign protein and about 3 weeks later measuring the thickness of both ears using a micrometer. The same protein is then injected into the left ear subcutaneously and saline injected in the right ear. Twenty-four hours later the ear thickness is again measured and the overall percentage increase in thickness is measured. This gives a reasonable qualitative measure of CMI. A more quantitative measure can be used by employing the migration of a radiolabel (^{125}I) into the ear after sensitization and measuring the radioactivity in the excised ear.

The other branch of the immune response involves the B cells, which are also stem cell in origin and mature in the bone marrow, liver, and gut and in the bursa of Fabricius in the chicken. They are primarily activated by bacteria and undergo further differentiation into mature B cells. They respond to further stimulation by proliferating and differentiating into memory B cells, which retain their ability to respond to antigen, and effector B cells or plasma cells which produce antibody. Antibodies are serum immunoglobulins (Ig) which have a wide range of specificity for different antigens. This arm of the response is known as humoral (circulating) immunity. Antibodies inactivate invaders by binding to them. There are five classes of immunoglobulins, IgG, IgA, IgM, IgD, and IgE. IgA predominates in bodily secretions and provides a primary defense

system because of its abundance in saliva, tears, colostrum, nasal mucosa, bronchial and vaginal secretions, etc. It exists in serum in both monomeric and polymeric forms and constitutes about 15% of serum Ig. IgM accounts for about 10% of serum Ig and normally exists as a pentamer. It is prominent in early immune responses to most antigens. About 75% of total Ig in the adult human is IgG. This monomeric antibody is produced later in the immune response than IgM. It activates the macrophage to ingest antigenic particles by coating them with antibody. IgE is a monomeric antibody which exists in only very small amounts in the blood. It stimulates mast cells to release histamine and heparin, which results in local blood vessel dilation and smooth muscle contraction. IgD is also monomeric and present in very minute amounts in blood. Its functions remain unclear. Both IgM and IgG are involved in complement fixation. Complement is a system of serum proteins which is the primary humoral mediator in antigen-antibody reactions.

A widely used assay for the assessment of humoral immunity is the Jerne plaque-forming cell assay. This assay gives a measure of antibody formation by single cells. Animals are given an injection of sheep red blood cells (the antigen). After a few days, spleen cells from immunized animals are plated out with sheep red blood cells on an agarose plate. When incubated, spleen cells release antibodies and red blood cells in the immediate vicinity are sensitized. In the presence of complement they lyse, leaving holes or plaques which can then be counted. This will detect IgM (i.e., early) antibody-forming cells and is known as the direct plaque-forming cell response. Indirect plaque-forming cells such as those forming IgG can be detected by overlaying the plate with the appropriate anti-immunoglobulin before adding complement.

T and B cell function are frequently assessed by the use of lymphocyte activation. This refers to an *in vitro* correlate of a process that regularly occurs *in vivo* when an antigen encounters a sensitized lymphocyte. Upon incubation with mitogens, usually plant lectins, many complex morphological and biochemical events occur. Early on there are membrane-related changes such as increased phospholipid turnover, fatty acyl exchanges, and increased permeability to cations. Shortly after the initial events RNA and protein synthesis occur, and then DNA synthesis. Any of these events can be measured to assess the ability of lymphocytes to respond to mitogen, but generally protein synthesis or DNA synthesis is measured. Protein synthesis can be measured by [^{14}C]leucine incorporation after 16–24 hours culture and DNA synthesis by [^3H]thymidine incorporation at 72 hours or later. The times at which these measurements are usually made and the culture conditions are important in later discussion. Specific mitogens can be used to activate transformation in T and B cells.

Phytohemagglutinin (PHA) and concanavalin A (Con A) stimulate T cells; pokeweed, T and B cells; and lipopolysaccharide, B cells.

Mitogen-induced activation of lymphocytes is nonspecific in that mitogens are polyclonal activators which stimulate a large proportion of cells. Specific activators like antigens stimulate only lymphocytes from sensitized animals. Antigen stimulus may be used instead of mitogen stimulus, but the animals must be presensitized to the antigen. A third possibility is to use stimulation by allogeneic lymphocytes, a procedure known as the mixed lymphocyte culture response (MLCR). All three procedures are used in studies of dietary effects on lymphocyte activation, although mitogen-induced activation is more common. This is perhaps the least informative assay so far as assessing the animals' immune response is concerned. Response to mitogen stimulation provides a measure of the functional capacity of lymphocytes to proliferate. What actually happens to lymphocytes *in vivo* in response to antigenic challenge may not be predictable from the results of mitogenic challenge. Mitogenic stimulation of circulating lymphocytes does provide a useful measure of immunocompetence in humans. In animal studies it is possible, and more informative, to use lymphocyte populations from more than one source. As will be seen, populations derived from different tissues do not necessarily exhibit the same responses.

The assays of immunocompetence discussed above are but some of the more commonly encountered assays in diet-immunity studies. There are many others, and some of these are listed in Table II.

Table II
COMMON METHODS USED IN THE ASSESSMENT OF THE IMMUNE RESPONSE^a

Assay	Cell and function assessed
Skin tests to recall antigens	T cells, delayed hypersensitivity
Binding of sheep RBC to form rosettes	Identification of human T lymphocytes
Lymphocyte activation	T and B cells, functional capacity to proliferate
Plaque-forming cell response	B cells, number of single cells producing antibody
Passive hemagglutination	B cells, estimate of antibody in serum
Lymphokine determinations	T cells, measure of mediators of cellular immunity
Surface Ig	Identification of B cells
Phagocytosis (various methods)	Macrophages, measure of phagocytic ability
Cytotoxicity (various methods)	Macrophages, measure of cytotoxic ability

^a These methods include only those mentioned in the text as having been used in studies of dietary fat and immunity. Numerous other methods exist.

B. MEMBRANE STRUCTURAL CHANGES AND BIOCHEMICAL EVENTS DURING LYMPHOCYTE ACTIVATION

As noted earlier the binding of ligand, whether it be mitogen or antigen, to a lymphocyte receptor is followed rapidly by biochemical events. There are changes in the composition of the membrane and the ordering of lipids in the membrane bilayers. Turnover of fatty acyl species is high, changes occur in the transport of ions and molecules across the membrane, and prostaglandins and other oxygenated fatty acids are produced with subsequent changes in cyclic nucleotides. Ultimately there is induction of RNA and protein synthesis followed by DNA synthesis and proliferation of cells.

Of particular significance in the present article are the dynamics of the phospholipids in the activated lymphocytes. Before considering the changes upon activation, it is useful to examine the fatty acid profiles of major phospholipids of immunocompetent cells. It is also of significance to see how changes in quality of dietary fat with respect to essential fatty acid (EFA) content alter these fatty acid profiles. Marshall and Johnston (1983) fed male weanling Sprague-Dawley rats purified diets containing either 10% of corn oil, soybean-linseed oil mixture, or linseed oil for a period of 8 weeks and determined the fatty acid profiles of several immunocompetent cell populations. Tables III and IV show some of these results. As the amount of dietary $18:3n-3$ relative to $18:2n-6$ increased, the $n-3$ fatty acid content of both choline and ethanolamine glycerophosphatides (CGP, EGP) increased and the $n-6$ fatty acid family decreased. The cell populations did, however, exhibit individual characteristics. Splenocytes and mast cells accumulated greater amounts of $22:6n-3$, whereas peripheral lymphocytes and thymocytes (results not shown) incorporated $20:5n-3$ as the major $n-3$ fatty acid. Rat thymocyte $20:4n-6$ concentration on the corn oil diet was about 25% higher than the other immune cells (Marshall and Johnston, 1983). Individual cell responses can be expected to lead to individual membrane structure and eicosanoid synthesis changes. Again, it cannot be assumed that an assay conducted on one cell type reflects the situation in all immune cell types.

It is established that the fatty acid profiles of lymphocytes are not only diet dependent but also individual in their response to diet. It should be emphasized that if isolated plasma membrane lipids are examined greater or less individual responses may be found. Nevertheless, the phospholipids examined are membrane-based and they give some measure of membrane changes. Fatty acid profiles of total lipids are less useful because they may include lipid droplets. The next question is, what happens to these fatty acid profiles when lymphocytes are activated? This question

Table III
THE EFFECT OF DIFFERENT DIETARY RATIOS OF α -LINOLENIC ACID TO LINOLEIC ACID ON PERIPHERAL LYMPHOCYTE AND SPLENOCYTE
ETHANOLAMINE PHOSPHOGLYCERIDE AND CHOLINE PHOSPHOGLYCERIDE FATTY ACID COMPOSITION^a

Fatty acid	Ethanalamine phosphoglyceride				Choline phosphoglyceride			
	Corn oil ^b	Soybean-linseed oil ^c	Linseed oil ^d	<i>p</i> ^e	Corn oil ^b	Soybean-linseed oil ^c	Linseed oil ^d	<i>p</i> ^f
Peripheral lymphocytes	(<i>n</i> = 3)	(<i>n</i> = 3)	(<i>n</i> = 3)		(<i>n</i> = 4)	(<i>n</i> = 2)	(<i>n</i> = 4)	
16:0	11.5 ± 0.9 ^f	15.5 ± 3.0	14.6 ± 0.9	NS ^g	56.7 ± 4.3	58.4 ± 0.2	58.7 ± 4.9	NS
18:0	16.9 ± 2.1	20.4 ± 2.5	16.9 ± 1.4	NS	10.0 ± 1.6	7.7 ± 0.6	9.2 ± 0.8	NS
18:1 ω 9	5.5 ± 1.0	6.5 ± 0.5	8.0 ± 0.6	NS	7.7 ± 1.1	6.9 ± 0.2	7.3 ± 0.8	NS
18:2 ω 6	6.6 ± 0.7	9.7 ± 0.8	11.6 ± 2.0	NS	10.6 ± 1.1	12.3 ± 0.4	8.8 ± 2.6	NS
18:3 ω 3	2.9 ± 1.2	2.6 ± 1.6	2.5 ± 0.5	NS	1.4 ± 0.4	1.8 ± 0.1	2.9 ± 0.4	NS
20:2 ω 6	4.4 ± 1.8	2.3 ± 1.3	2.1 ± 1.0	NS	2.3 ± 0.6	1.7 ± 0.2	1.5 ± 0.3	NS
20:3 ω 6	2.3 ± 0.1	2.1 ± 0.3	2.1 ± 0.1	NS	0.7 ± 0.2	1.0 ± 0.3	0.9 ± 0.1	NS
20:4 ω 6	33.2 ± 1.9	26.9 ± 3.7	21.4 ± 2.2	<i>p</i> < 0.05	12.4 ± 1.7	5.9 ± 0.05	4.5 ± 0.9	<i>p</i> < 0.01
20:5 ω 3	ND ^h	2.9 ± 0.3	8.7 ± 0.8	<i>p</i> < 0.005	0.3 ± 0.3	1.6 ± 0.6	2.4 ± 0.6	<i>p</i> < 0.005
20:4 ω 6	14.2 ± 1.4	5.8 ± 0.8	2.3 ± 0.5	<i>p</i> < 0.005	2.3 ± 0.3	1.3 ± 0.7	1.1 ± 0.6	NS
22:5 ω 3	ND	4.1 ± 0.8	4.7 ± 0.9	<i>p</i> < 0.005	0.3 ± 0.3	0.8 ± 0.2	1.0 ± 0.2	<i>p</i> < 0.005
22:6 ω 3	ND	0.6 ± 0.4	0.3 ± 0.2	NS	0.4 ± 0.2	0.8 ± 0.8	0.4 ± 0.4	NS
<i>n</i> = 6/ <i>n</i> = 3 ⁱ	20.8	4.6	2.4		11.7	4.4	2.5	

Splenocytes	(n = 4)	(n = 5)	(n = 6)	(n = 4)	(n = 5)	(n = 3)
16:0	13.2 ± 0.9	12.3 ± 1.2	11.1 ± 0.9	37.6 ± 2.8	36.9 ± 1.2	39.1 ± 3.0
18:0	20.8 ± 0.7	19.2 ± 1.9	16.7 ± 0.8	8.5 ± 1.1	8.2 ± 0.6	7.8 ± 0.3
18:1 ω 9	7.3 ± 1.0	7.8 ± 0.2	9.6 ± 0.3	9.7 ± 1.3	10.8 ± 0.8	13.5 ± 1.0
18:2 ω 6	10.4 ± 1.7	9.6 ± 0.6	11.6 ± 0.8	17.0 ± 1.0	18.0 ± 1.8	16.3 ± 1.6
18:3 ω 3	1.2 ± 0.3	2.3 ± 0.6	3.5 ± 0.8	0.5 ± 0.5	2.0 ± 0.5	4.7 ± 0.6
20:2 ω 6	2.2 ± 0.2	1.9 ± 0.6	1.4 ± 0.1	4.7 ± 0.3	2.7 ± 0.1	1.8 ± 0.4
20:3 ω 6	2.5 ± 0.5	2.0 ± 0.3	4.0 ± 0.8	1.7 ± 0.2	1.6 ± 0.3	2.3 ± 1.3
20:4 ω 6	29.6 ± 1.4	28.9 ± 2.2	21.8 ± 0.7	15.7 ± 1.6	11.1 ± 1.5	9.3 ± 0.9
20:5 ω 3	ND	3.0 ± 0.3	6.5 ± 0.3	ND	1.6 ± 1.1	3.1 ± 1.0
22:4 ω 6	8.1 ± 0.4	2.7 ± 0.2	1.7 ± 0.3	1.5 ± 0.3	1.8 ± 0.3	0.2 ± 0.2
22:5 ω 3	0.9 ± 0.2	7.0 ± 0.7	8.2 ± 0.3	0.4 ± 0.2	1.7 ± 0.1	1.1 ± 0.1
22:6 ω 3	1.3 ± 0.2	2.5 ± 0.3	2.3 ± 0.3	0.4 ± 0.1	1.2 ± 0.3	0.5 ± 0.1
n - 6/n - 3	15.4	3.1	1.9	32.0	5.4	3.2

^a Reprinted with permission from Marshall, L. A., and Johnston, P. V. (1983). *Lipids* 18, 737-742. Copyright American Oil Chemists' Society.

^b 10% corn oil.

^c 10% soybean-linseed-oil mixture.

^d 10% linseed oil.

^e *p* Values are a result of analysis of variance (ANOVA).

^f Mean ± standard error of the mean (SEM).

^g NS = not significant.

^h ND = not detectable.

ⁱ n - 6/n - 3 = total n - 6/total n - 3 fatty acids.

Table IV

THE EFFECT OF DIFFERENT DIETARY RATIOS OF α -LINOLENIC ACID TO LINOLEIC ACID ON RAT MAST CELL ETHANOLAMINE PHOSPHOGLYCERIDE AND CHOLINE PHOSPHOGLYCERIDE FATTY ACID COMPOSITION^a

Fatty acid	Ethanolamine phosphoglyceride			Choline phosphoglyceride		
	Corn oil ^b (n = 5)	Linseed oil ^c (n = 5)	p ^d (% of total)	Corn oil ^b (n = 5)	Linseed oil ^c (n = 5)	p ^d (% of total)
16:0	9.9 ± 0.9 ^e	10.7 ± 0.6	NS ^f	23.1 ± 2.6	15.8 ± 1.5	NS
16:1 ω 7	1.7 ± 0.3	1.7 ± 0.4	NS	5.6 ± 0.7	6.8 ± 0.6	NS
18:0	14.0 ± 1.6	14.1 ± 1.6	NS	9.6 ± 0.8	12.0 ± 0.9	NS
18:1 ω 9	10.6 ± 0.5	13.3 ± 0.8	NS	11.5 ± 0.4	15.5 ± 0.2	0.001
18:2 ω 6	12.2 ± 1.2	11.6 ± 1.4	NS	16.4 ± 0.5	17.4 ± 1.1	NS
18:3 ω 3	ND ^g	2.3 ± 0.4	0.012	ND	4.0 ± 0.3	0.001
20:4 ω 6	31.8 ± 2.5	20.2 ± 0.9	0.016	22.7 ± 2.1	12.1 ± 0.7	0.009
20:5 ω 3	0.3 ± 0.2	9.5 ± 1.1	0.002	0.5 ± 0.3	5.6 ± 0.6	0.003
22:4 ω 6	10.2 ± 0.7	2.3 ± 0.4	0.001	3.8 ± 0.3	1.5 ± 0.6	0.018
22:5 ω 6	2.5 ± 0.2	0.4 ± 0.3	0.021	0.8 ± 0.1	0.2 ± 0.1	0.002
22:5 ω 3	0.6 ± 0.2	9.0 ± 0.8	0.004	0.3 ± 0.0	4.5 ± 0.5	0.001
22:6 ω 3	1.3 ± 0.3	3.0 ± 0.4	0.087	0.5 ± 0.1	1.1 ± 0.2	0.011
Others	2.4	3.1		4.7	3.0	
n = 6/n = 3 ^h	45.2	1.6		33.6	2.1	

^a Reprinted with permission from Marshall, L. A., and Johnston, P. V. (1983). *Lipids* **18**, 737-742. Copyright American Oil Chemist's Society.

^b 10% corn oil.

^c 10% linseed oil.

^d p values are a result of ANOVA.

^e Mean ± standard error of the mean (SEM).

^f NS = not significant.

^g ND = not detectable.

^h n = 6/n = 3 = total n = 6/total n = 3 fatty acids.

has been reviewed several times over the past few years (Ferber and Resch, 1977; Meade and Mertin, 1978; Ferber *et al.*, 1980) and here will be summarized.

During the early phase of the activation process there is a preferential transfer of long-chain fatty acids from intracellular sources to phospholipids of the plasma membrane. *De novo* fatty acid synthesis is low but turnover is high. The scheme for the mechanism of turnover is shown in Fig. 1. In lymphocytes the acyl-CoA:1-acyl-sn-glycero-3-phosphocholine O-acyltransferase (acyl-CoA:lysolecithin acyltransferase, EC 2.3.1.23) appears to be located primarily in the plasma membrane. Upon binding of

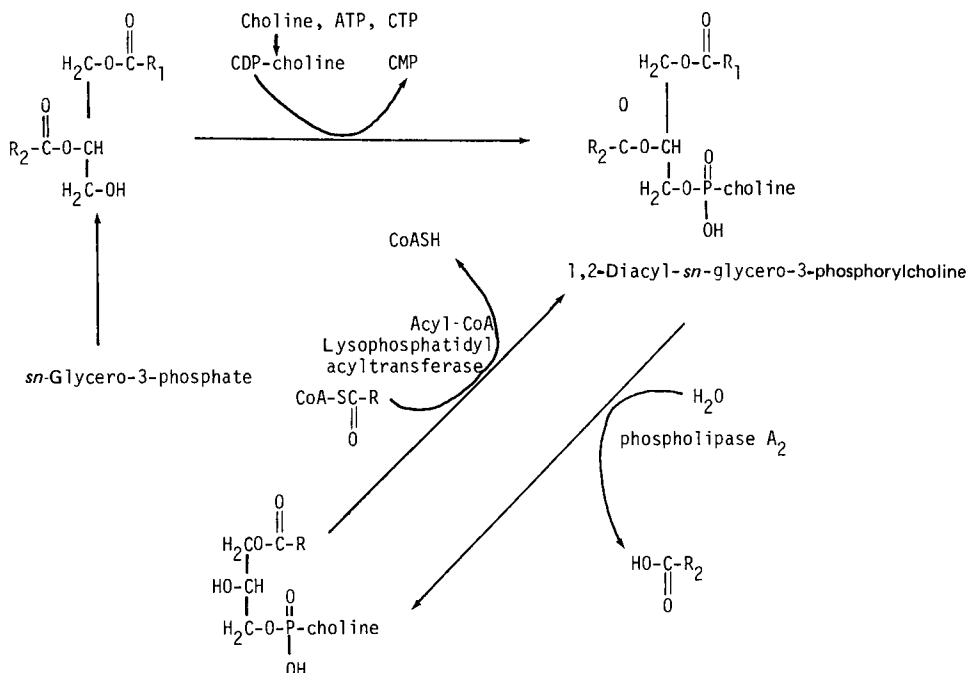


FIG. 1. Metabolism of choline glycerophosphatide.

ligand the enzyme is activated. It has a high affinity for polyunsaturated fatty acids such as arachidonic acid ($20:4n-6$) (Ferber *et al.*, 1980). As proof that the activation of the enzyme occurs immediately upon binding of the ligand, Ferber *et al.* (1980) note that the process does not require metabolic energy as it takes place to the same degree at 0°C . However, as Meade and Mertin (1978) pointed out, there are differences in the kinetics of Con A binding and activation of the acyltransferase. With increase in Con A concentration its binding increases with hyperbolic kinetics associated with a simple, saturable receptor. The activation of the acyltransferases, on the other hand, follows sigmoid kinetics. This does not, however, preclude that the activation of the acyltransferase is the direct result of Con A binding.

Lectins that stimulate transformation also increase turnover of the inositol and phosphate moieties of phosphatidylinositol (PI), and this occurs in both T and B cells (Maino *et al.*, 1975). There are lectins which can cause certain phenomena like capping and endocytosis of receptors (redistribution of receptors) but do not initiate transformation or stimulate increased PI turnover. This is now generally believed to be due to the fact

that lectins initiating transformation must cross-link, or aggregate, receptors.

How the ligand-induced changes in membrane lipid composition and, specifically in unsaturation, trigger the rest of the biochemical events has been the subject of much speculation. Often the term "increased fluidity" is used to explain the activation of other enzymes, changed ion movements, etc. "Fluidity" is most frequently measured by fluorescence polarization or spin labeling. There are problems with both procedures and their use for assessing an average "fluidity" of membranes has frequently been criticized. As Schreier *et al.* (1978) point out, most membrane environments are not isotropic, and the use of the term "fluidity" implies the reciprocal of viscosity. They further cite examples of "fluidity" being measured using only order parameters and note that there is not necessarily a correlation between rates of molecular motion and degree of molecular order. Some of the problems associated with the very complex question regarding whether or not lipid compositional changes can affect membrane structure, "fluidity," and signal transmission were addressed by Traill and Wick (1984). They concluded that changes in local lipid environments may be expected to affect both binding and cross-linking events at receptors. They propose that, after the increase in phospholipid fatty acid unsaturation is induced by the acyltransferase, a wave of phospholipid "fluidity" increase occurs along the membrane which propagates the signal and activates other enzymes which catalyze the sequence of events that follow. That lipid unsaturation is crucial to the activation of certain membranes during lymphocyte activation events was also concluded by Ferber *et al.* (1980). They showed that the incorporation of both oleoyl CoA and arachidonyl CoA resulted in a continuous and nearly identical decrease in overall membrane microviscosity as measured by fluorescence polarization. However, the activity of $\text{Na}^+ - \text{K}^+$ -ATPase was elevated only when an optimal amount of arachidonic acid was incorporated.

At this time it is probably best to conclude only that there is evidence that the lipid composition of the membrane does influence ligand binding and propagation of events in lymphocyte activation. However, overall microviscosity changes may not have as much meaning as the local lipid environment of specific enzymes. Moreover, the changes required to initiate individual enzyme activities may be specific.

C. ROLE OF PHAGOCYTES

There are two major classes of phagocytes in man, the circulating phagocytes which includes monocytes and granulocytes (neutrophils, eo-

sinophils, and basophils), and the fixed phagocytes which are mainly macrophages. Neutrophils, the predominant granulocyte in man, are also known as polymorphonuclear leukocytes (PMN). All of the phagocytes arise from the bone marrow. The macrophages mainly reside in the peritoneum, liver, spleen, lymph nodes, bone marrow, and lungs. Monocytes and macrophages are involved in phagocytosis and killing of microorganisms, removal of debris and dead cells, regulatory interactions in hematopoiesis, synthesis of biologically active compounds including eicosanoids, and cooperative and effector functions in the immune response. These last two functions are of major interest in this article, and we shall mainly be concerned with the macrophage.

As noted above, quite a lot of information exists about the way in which the remodeling of the lymphocyte plasma membrane takes place. The early activation of lysolecithin acyltransferase after the mitogenic stimulus explains, at least in part, the changes in the acyl chains. The effects of the oxygenated products of arachidonic acid, the prostaglandins (PG), and related eicosanoids have also been studied. The effects are very complex. The complexity arises from the fact that PGs have been found to have both stimulatory and suppressive effects on immune responses. This question is discussed more fully in Section II,D. One major problem is that there is uncertainty regarding the population of cells producing PG, particularly when a typical peripheral leukocyte population is used. Such preparations may contain T and B cells, monocytes, and null lymphocytes, and in some cases platelet contamination is considerable. Platelets can particularly confuse matters as they are prolific producers of PG and other eicosanoids. It is, however, generally now agreed that among immunocompetent cells macrophages are the major source of PG and thromboxanes (Parker, 1981; Goldyne and Stobo, 1981). Synthesis by T and B lymphocytes cannot, however, be ruled out. It is known that mitogenic stimulation requires the presence of small numbers of macrophages (Habu and Raff, 1977; Taniguchi *et al.*, 1977). Therefore, in addition to considering fatty acid metabolism in lymphocytes, we also need to consider related events in the macrophage.

While the reactions taking place in the reformation of phosphoglycerides in the lymphocyte are quite well understood, less is known about the degradation part of the cycle which provides the lysophosphoglycerides for acyl exchange. It has been proposed that macrophages are the exogenous source of the lysophosphoglycerides (Munder *et al.*, 1969). This knowledge led to studies on fatty acid metabolism in macrophages. By judicious use of stimulators and inhibitors and [^{14}C]20:4 $n-6$, it has been shown that macrophages stimulated by zymosan continuously degrade phosphoglycerides only when the acyltransferase is inhibited by an

agent such as *p*-chloromercuribenzoate (Meerphol *et al.*, 1976). Thus it appears that the amount of free 20:4 n - 6 is controlled by the high acyltransferase in macrophages as well as by its phospholipase A₂ (EC 3.1.1.4). With a phospholipase activator, Ca²⁺ ionophore A 23187, CGP is degraded without the *p*-chloromercuribenzoate because the ionophore itself inhibits the acyltransferase. Zymosan stimulation does not increase release of 20:4 n - 6 because phagocytosis of inert latex particles has no effect on the reaction. The possibility exists, therefore, for the activation of two distinct aspects of phospholipid metabolism of two different cell populations. For example, Con A will activate the acylation reaction in lymphocytes but have no effect on phospholipase and none on the de- and reacylation in macrophages. Zymosan will selectively activate macrophage phospholipase but have no effect on acylation. Thus it seems reasonable to propose that macrophages may provide lyso compounds and free 20:4 n - 6 to other cells. This controlled availability of free 20:4 n - 6 has implications for eicosanoid synthesis. There is also an alternate pathway for release of 20:4 n - 6 by lymphocytes. This is the phospholipase C (EC 3.1.4.3) and diacylglycerol lipase (EC 3.1.1.34) pathway. Lymphocytes have a highly active cytosolic phospholipase C for phosphatidyl inositol (PI). Parker and co-workers (Parker, 1981) have conducted studies on this aspect. Some evidence suggests that lectin-stimulated peripheral blood lymphocytes do not exhibit increased diacylglycerol [¹⁴C]20:4 n - 6 labeling early in the response. This is in contrast to the substantial increases in the lipid in mast cells within seconds (Kennerly *et al.*, 1979). This same group of researchers has studied a so-called thromboxane (TXA₂) effect using inhibitors of thromboxane synthesis. Making use of the nonmetabolizable aminoisobutyric acid (AIB) to study the alanine-preferring transport system, they have shown that inhibitors of TXA₂ synthesis interfere with lectin-stimulated increases in AIB transport in lymphocytes (Parker, 1981). They have also examined the effects of lipoxygenase products and have proposed that, while there is a low overall hydroxylated fatty acid incorporation into lymphocyte cellular membrane fractions, the local incorporation of such fatty acids could markedly increase membrane microviscosity (Parker, 1981) and have effects on membrane function. The role of fatty acids such as 15-hydroxyeicosatetraenoic acid (15-HETE) does, however, remain unclear. The 5,12-diHETE has been shown to produce a rapid increase in Ca²⁺ exchange in polymorphonuclear leukocytes (Naccache *et al.*, 1981).

Several points emerge from the above discussion and may be summarized as follows:

1. Small numbers of macrophages are essential for mitogen stimulation of lymphocytes.

2. Lyso compounds may be transferred from macrophages to lymphocytes and alter structure and function. As ionophores they may in fact be directly involved in lymphocyte activation.

3. Macrophages, but not lymphocytes, have significant phospholipase activity.

4. Reacylation reactions are initiated in the lymphocyte directly after binding and lead to membranes of higher unsaturation. These changes result in an overall change in microviscosity, but more specific local alterations of membrane enzyme activities are involved.

5. Lysophosphatide availability and free fatty acid (notably $20:4n-6$), essential to the unsaturation changes, are controlled by the acyltransferase activity. The $20:4n-6$ availability has implications in PG and other eicosanoid metabolism.

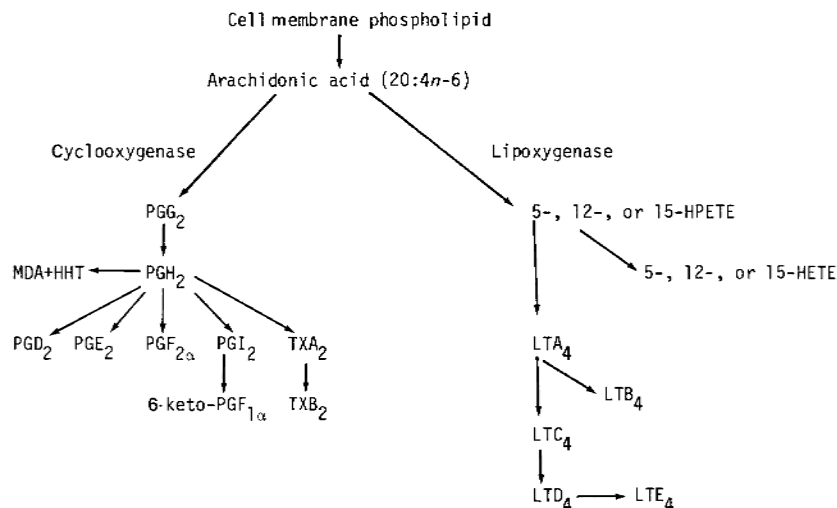
6. Prostaglandins are known modulators of the immune response.

7. Thromboxanes and lipoxygenase products may be involved in modulating changes in activation events.

D. EICOSANOID MODULATION OF IMMUNE RESPONSES

Under normal circumstances the major eicosanoid precursor in most cells is $20:4n-6$, which gives rise to the 2-series PG, that is, those having two double bonds in the side chains. The other precursors, $20:3n-6$ and $20:5n-3$, give rise to the 1- and 3-series PG, respectively. Once $20:4n-6$ is released by the phospholipase, it may enter the PG synthase (EC 1.14.99.1) and/or lipoxygenase (EC 1.13.11.12) pathways as shown in Fig. 2. Cyclooxygenase products include the relatively stable PGE_2 , PGI_2 or prostacyclin, and the thromboxanes (TX). The lipoxygenase pathway gives rise to the hydroperoxy- and hydroxyeicosatetraenoic acids and to the more recently discovered leukotrienes (LT). Comparable products of both pathways may arise from the other eicosanoid precursors, $20:3n-6$ and $20:5n-3$.

The PGs of the 2-series and lipoxygenase products of $20:4n-6$ have been the most extensively studied and there is considerable information on 1-series PG, especially PGE_1 . Controversy exists regarding whether or not there is *in vivo* production of 3-series PG. Both $20:3n-9$ and $20:5n-3$ appear to be poor substrates for human platelet PG synthase. Several workers have reported that the cyclooxygenase of sheep vesicular gland does not oxygenate $20:5n-3$ unless peroxides are present (Falardeau *et al.*, 1976; Lagarde *et al.*, 1977; Needleman *et al.*, 1979; Culp *et al.*, 1979). This implies that 3-series PGs reported in tissue homogenates are artifacts of the homogenization aeration effects. However, Boukhchace and Lagarde (1982) showed that $20:5n-3$ can be efficiently utilized by the cyclooxygenase provided that $20:3n-6$ and $20:4n-6$ are present.



PGG₂, PGH₂, PG endoperoxides

PGI₂, prostacyclin

HHT, hydroxyheptadecatrienoic acid

HPETE, hydroperoxyeicosatetraenoic acid

HETE, hydroxyeicosatetraenoic acid

LT, leukotriene

TX, thromboxane

MDA, malondialdehyde

FIG. 2. Oxygenated metabolites of arachidonic acid.

They believe that the effect is mediated via the $n - 6$ series hydroperoxides or peroxides. More recently Fischer and Weber (1984) showed that after the ingestion of either cod liver oil or mackerel (4–15 g 20:5 $n - 3$ /day) human volunteers produced PGI₃, as evidenced by the occurrence of Δ^{17} -2,3-di(nor-6-ketoprostaglandin)F_{1 α} (PGI₃-M) in urine. PGI₃-M is analogous to PGI₂-M, which has been shown to be the major urinary metabolite of PGI₂ (Falardeau *et al.*, 1981). These studies are significant since they suggest that 3-series PG may be synthesized by many cells if considerable 20:5 $n - 3$ is present in the presence of 20:4 $n - 6$ and 20:3 $n - 6$. The PG of different series may exhibit different physiological effects. For example, TXA₃ in contrast to TXA₂ does not aggregate platelets (Needleman *et al.*, 1976, 1979). Thus, shifts in the production of the trienoic series relative to the dienoic may markedly influence cell function.

Which cells of the immune system synthesize eicosanoids is still a matter of controversy. As noted earlier, the cell populations used are frequently heterogeneous. While one cell type may predominate, most work has been conducted using cells contaminated by other cell types. A

major area of controversy concerns whether or not lymphocytes produce PG. Parker (1981) has reviewed arachidonic acid metabolism by both the cyclooxygenase and lipoxygenase systems in activated lymphocytes. Goodwin and Webb (1980) have given a comprehensive review of the regulation of the immune response by PG. They have given concise illustrations regarding the apparent role of PG in immunity. They showed that a population of glass-adherent T lymphocytes (60% B cells, 40% T cells) are activated by prostaglandin E_1 or E_2 (PGE_1 or PGE_2) to become suppressor cells. Either a mitogenic or an antigenic stimulus causes release of PG by lymphocytes or macrophages, or both, which activates the T suppressors to produce a non-PG molecule which suppresses blastogenesis in both T and B cells. PGs are therefore directly suppressive and can also activate T suppressors to control blastogenesis. They further note that there appears to be both a PG-dependent and a PG-independent suppressor cell population. Upon antigenic stimulation, PGs activate nonspecific suppressor cells which may exist together with antigen-specific cells. The authors give the general rule that early and primary events of the immune response are sensitive to inhibitory mechanisms involving PG. In later stages and secondary responses, on the other hand, PG shifts towards the modulation of secretory functions of both T cells and B cells. As these authors see it, therefore, PGs are involved in all phases of regulation of immunoresponsiveness.

Other authors give a very different picture of PG involvement in immunity. Goldyne and Stobo (1981) reviewed the evidence for PG and TX synthesis by macrophages. There appears little controversy here. All authors appear to agree that macrophages are major producers of PG, TX, and, depending upon their source, lipoxygenase products including leukotrienes (LT). Goldyne and Stobo point out that while macrophages without exception exhibit active cyclooxygenase, the studies of its presence in lymphocytes provide equivocal data. Their review of studies on lymphocyte PG concluded with the statement that among immunocompetent cells macrophages are the primary and possibly the sole source of PG and TX. However, they allow that a significant macrophage-lymphocyte interaction may occur which involves utilization by macrophages of $20:4n-6$ liberated by activated lymphocytes. Thus they envision three possible interactions between macrophages and lymphocytes. The first involves direct stimulation of macrophage PGE_2 synthesis, the second indirect stimulation of macrophage PGE_2 synthesis by a T cell-derived lymphokine, and the third indirect stimulation of macrophage PGE_2 synthesis by free $20:4n-6$ released from "non-PG-producing T cells" by stimuli such as PHA. This concept has one particularly attractive aspect in that it provides an explanation for the essential role of the macrophage in the

regulation of some T cell activities. T cells and macrophages become interdependent because the macrophage provides PG for stimulus of the T cell to produce lymphokine or to produce free $20:4n-6$ for the macrophage to produce PG. Whether or not lymphocytes produce PG and related products does, however, remain unclear. Goldyne and Stobo (1981) strongly believe that the macrophage is the sole source of PG, and they feel maintaining the concept that all cells produce PG may obscure their role in immunologic and other functions. Unequivocal results pertaining to this concept are, however, lacking. At this stage the situation must be summarized with caution. The accumulated evidence indicates that immune cells synthesize eicosanoids. In the case of the macrophage the evidence is compelling but less so in the case of lymphocytes. Reports of changed responses of cells associated with changes in PG synthesis probably represent the overall result of direct and indirect PG effects and interactions between different cell types in the population.

That PG and related metabolites influence immune responses does not appear to be in doubt. Again, because of methodological problems, the literature is complex and difficult to interpret. There have been two major approaches to the study of PG effects on immune function. The first involves the addition of exogenous PG to cell preparations and measurement of various activities such as the effect of produced lymphokines on macrophage migration, lymphocyte transformation, and PI turnover. The second approach is the use of inhibitors of PG synthesis, indomethacin in particular. Goldyne and Stobo (1981) reviewed these studies extensively.

The addition of both exogenous PG and inhibitors presents pitfalls. One of the most common errors has been the addition of PG at levels much higher than they are normally produced. A range of 10^{-9} – 10^{-7} M can be considered a reasonable physiological dose. In 1977 Goodwin *et al.* pointed out the disparity between the amount of prostaglandin E (PGE) produced during mitogen stimulation (10^{-8} M) and the amount used by investigators to demonstrate that PGE depressed mitogen stimulation (10^{-5} M). Other problems are that investigators use different levels of mitogen (and often only one) and different concentrations of serum in the culture medium. Variations in response to different levels of mitogen and the binding of PG to serum proteins are no doubt responsible for the wide variation in results reported. The use of the inhibitor indomethacin leads to other difficulties. Indomethacin has effects on metabolism other than its inhibition of PG synthesis. Moreover, when the cyclooxygenase is inhibited the lipoxygenase could be more active. Thus effects attributed to the blockage of PG synthesis may well be the result of a change in the eicosanoid profile. Indeed, this is one of the major problems in the area of

eicosanoids and immunity, as usually only one or two PG have been determined, making the literature very difficult to interpret.

Again the summary must be cautious. The *in vitro* studies, in spite of their flaws, collectively indicate that PG and other eicosanoids modulate immune cell activities, but precise mechanisms are not known. The statement made by Goldyne and Stobo at the end of their exhaustive critical review stands as well today as it did 3 years ago. These authors state, "The role of a given PG, such as PGE₂, may only be specifically defined in relation to a given cell at a given time."

III. Dietary Fat and the Immune Responses

Interest in the immunological effects of lipids emerged in the last decade. By 1978 there was sufficient information for the first comprehensive review on the subject by Meade and Mertin, although aspects of it had been reviewed earlier by Ferber and Resch (1977). At the time of the Meade and Mertin (1978) review there had been relatively few dietary studies in which attempts had been made to influence immune functions by changes in dietary fat. This section emphasizes the dietary studies reported since that review.

A. QUALITY AND QUANTITY OF FAT

Observations that polyunsaturated fatty acids (PUFA) prolong allograft rejection (Ring *et al.*, 1974), decrease the severity of an inducible autoimmune disease (Clausen and Møller, 1969; Selivonchick and Johnston, 1975), and increase tumor incidence and growth (Carroll and Khor, 1975) were among several that encouraged investigators to study the immune response in animals fed diets differing in PUFA and saturated fat (SF) content and in total fat content. Erickson *et al.* (1980) fed female C57BL/6J mice semipurified diets containing 2, 8, 16, or 24% coconut or safflower oil with casein adjusted to provide 20% of the total metabolizable energy. The mice were mated and the progeny were weaned to the same diets as their dams. Pups were sampled 2, 4, and 6 weeks after birth and both humoral and CMI responses were measured.

The serum immunoglobulins IgG₁ and IgG₂ but not IgM or IgA increased in mice fed PUFA as compared to SF. The percentage of immunoglobulin positive (Ig⁺) cells was changed by diet but not the number of T cells. Neither quantity nor degree of saturation altered lymphocyte blastogenesis in response to the B-cell mitogen lipopolysaccharide (LPS).

High PUFA suppressed T-cell blastogenesis and low levels or a deficiency of PUFA enhanced the response. The particular changes seen varied with the development of the immune responses. For example, in 2-week-old mice saturation of fat increased splenic weight, whereas at 6 weeks fat concentration influenced it.

The authors explained the changes in serum Ig levels as probably due to diet-induced changes in the liver, which is the major source of B cell precursors from day 10 of gestation until birth. To explain the changes in T cell responses, the authors favored the idea that they were due to changes induced by PUFA in the fluidity of the plasmalemma. This is based on the observations of Ferber and Resch (1977), who showed that when the lymphocyte is activated there is a marked turnover in membrane fatty acids, with increased incorporation of PUFA. The Erickson *et al.* study is difficult to interpret for several reasons. The SF diet is an EFA-deficient diet; thus, as the authors state, they were observing not only a SF versus PUFA effect but also an EFA-deficient versus an EFA-replete effect. No fatty acid analyses of the lymphoid tissues and cells were conducted. Although changes can be expected, the extent of them is unknown. Another major problem with the mitogenesis studies is that the assay was conducted using a medium containing 10% fetal bovine serum (FBS). A total culture period of 72 hours was used. The same error was made in an earlier report by these workers (Erickson *et al.*, 1979). It was pointed out a few years ago that cells in culture will adopt the fatty acid profile of the medium in which they are cultured (King and Spector, 1981). However, as will be seen, this is a common flaw in many studies in this field. We became concerned about this and conducted studies to determine to what extent splenocytes would adopt the fatty acid profile of the serum in the medium in which they were subjected to relatively short-term culture. To determine whether or not such changes would result in change in a cell function, we measured the PG synthesis by the cultured splenocytes. It was found that splenocyte phosphoglycerides adopted a fatty acid profile similar to the culture serum within a 48-hour period, and this was accompanied by changes in PG synthesis which reflected the new fatty acid patterns (Loomis *et al.*, 1983). Thus, when cells from dietary studies are cultured in FBS the dietary effects may be abrogated or masked. It explains why DeWille *et al.* (1981a), found no effect of EFA deficiency on lymphocyte blastogenesis. In their study the cells were cultured in 5% FBS for 108 hours. Gurr (1983) pointed out another problem with this study. He noted that the results of the [³H]thymidine incorporation as a measure of blastogenesis were given as cpm rather than as a stimulation index (SI) which relates radioactivity incorporated in the presence of mitogen to radioactivity incorporated in the absence of mito-

gen. Calculation of the SI revealed that there was no difference between lymphocyte transformations of PUFA and SFA in 2-week-old animals, whereas in 4-week-old animals the PUFA group lymphocytes were more stimulated and in 6-week-old animals less stimulated.

In another report from Erickson and co-workers (Erickson *et al.*, 1983) some of the earlier problems were resolved. In this study six dietary groups were used; one group was fat-free and therefore EFA deficient, the other contained minimal EFA (1.4% en) and either 8 or 20% safflower oil, or 8 or 20% coconut oil. The diets were adjusted to be equal in percent protein and carbohydrate, but no adjustment was made for vitamins and minerals. Mice (C57BL/6J, BALB/cAnN, and C3H/HeJ) were fed the diets for 4 weeks before being immunized. Blastogenesis of lymphocytes from dietarily manipulated mice in response to alloantigens (spleen cells) from control mice was examined. The results of this experiment are summarized in Table V.

Lymphocyte transformation from dietarily manipulated mice was significantly greater for mice fed the EFA-deficient diet, with the exception of the 20% PUFA diet. In an additional experiment, Erickson *et al.* showed that if dietarily manipulated mice (EFA 1.4% en) acted as responders to stimulators from mice fed a stock diet the EFA deficiency depressed the blastogenic response. When additional fat was added to the diet, the blastogenic response decreased. When dietarily manipulated

Table V
LYMPHOCYTE TRANSFORMATION IN DIETARILY
MANIPULATED MICE TO ALLOANTIGEN FROM
CONTROL MICE

Dietary fat (%)	[³ H]Thymidine ^a (cpm)
0	52,500 ± 1,740 ^b
0.5 EFA ^c	26,500 ± 5,000
8 PUFA ^d	23,600 ± 4,000
8 SFA ^e	28,000 ± 6,000
20 PUFA	42,000 ± 4,100
20 SFA	25,700 ± 3,000

^a C57BL/6J mouse stimulator cells for responding BALB/c mice fed a stock diet. EFA = essential fatty acid; PUFA = polyunsaturated fatty acid; SFA = saturated fatty acid.

^b Mean ± SEM.

^c EFA control.

^d Safflower oil + 1.4% en EFA.

^e Coconut oil ± 1.4% en EFA.

cells acted as stimulators and the responders were from the stock diet group, blastogenesis was more influenced by concentration than type of fat. Again, the meaning of these findings is difficult to interpret because the mixed lymphocyte culture required 120 hours. Although the authors show fatty acid profiles of lymphocytes from mice on all the diets, these may not reflect the fatty acid patterns existing when the cultures were labeled with [^3H]thymidine at 108 hours. The authors themselves question the validity of this part of the study.

In addition to the transformation studies, the effect of dietary fat on peritoneal exudate cell (PEC)-mediated cytotoxicity toward a P51 melanoma target cell was measured by radioisotope (^{51}Cr) release. The cytotoxicity results are amenable to interpretation because only 4 hours of culture were required. Changes in the diet-induced fatty acid profile are less likely to have been changed by such a short culture. The results showed that cytotoxicity was not influenced by saturation of fat at high concentration, but when the concentration of fat was reduced mice fed the PUFA had suppressed cytotoxicity when compared to mice fed SFA.

In contrast to the above reports, Clifford *et al.* (1983) concluded that mitogen-induced lymphocyte transformation was unrelated to the saturation of dietary fat and was associated negatively and weakly with the amount of dietary fat absorbed. These investigators fed weanling rats for 23 days casein-based diets containing purified and mixed triglycerides. In addition to lymphocyte transformation to PHA, Con A, and pokeweed mitogen (PWM), lymphoid organ weights and fatty acid profiles of the total lipid from plasma, spleen, and thymus were determined. Saturated, even-numbered carbon fatty acid series from C4 to C18, 18:n - 9 *cis*, 18:1n - 9 *trans*, and 18:2n - 6 *cis,cis* were the constituents of the triglycerides. The mixed triglycerides were corn oil and partially hydrogenated soybean oil. The diets were supplemented with 0.82 g of safflower oil/100 g to provide EFA. Digestibility and absorption of protein, fat, and calories were determined. With the exception of tristearin, tripalmitin, and trimyristin, the digestibility coefficients for all triglycerides were 90% or greater.

Since a serum-free medium was used for the lymphocyte activation assay, the results of this study should reflect the dietary treatments. With the exception of the poorly digested fats, the fatty acid profiles of the plasma, spleen, and thymus reflected the dietary fat. The profiles of the tissues, however, show a striking similarity to each other. This is to be expected when the total lipid is examined because the greater contribution to the fatty acid profile is made by the triglycerides which, compared to the membrane-based phosphoglycerides, rapidly reflect the dietary fat.

The thymus is very rich in triglycerides, and much of the spleen total lipid arises from blood in the dense vascular bed of the organ.

It is difficult to concur with the overall conclusions drawn by the authors of this study that mitogen-induced lymphocyte transformation need not necessarily be altered when fatty acid profiles of plasma, spleen, and thymus are altered. They used a relatively short feeding period of 3 weeks, and the fatty acid profiles probably largely reflect changes in blood lipids and triglyceride fractions. Moreover, the gas-chromatographic column used (6 meter) had an OV 275 packing which gives poor recovery of PUFA and renders the column of doubtful accuracy when complex fatty acid mixtures are analyzed (Walker, 1981).

Another recent study on the effect of quality and quantity of dietary fat on the immune system was reported by Lockniskar *et al.* (1983). Sprague-Dawley rats were fed at a 5% level (control diet) or a specific fat such as saturated (beef tallow), PUFA (corn oil), or partially saturated (Crisco) at 24% level. All diets were balanced with respect to the nutrient:calorie ratio. The saturated-to-unsaturated ratios were as follows: mixed fat, 28:72; beef tallow, 44:56; corn oil, 14:82; partially saturated fat (PSF), 23:77; respectively. The beef tallow and Crisco would contain marginal EFA. The Crisco would contain trans isomers although not the 20% as trielaidin as stated by the authors, which must be an error. Rats were fed the diets for 2.5 or 5 months. Lymphocyte response to PHA, Con A, and PWM was determined. In a 2.5-month feeding experiment the assay was conducted on both splenocytes and mesenteric lymph nodes (MLN) and the assays were conducted in rat autologous serum, allogeneic serum from a different dietary group, or FBS, all at 10%. In a second experiment rats were fed for 5 months and the assay conducted on splenocytes and MLN using FBS-supplemented serum. Spleens and MLN from the first experiment were also examined histopathologically.

The mitogens used in this study differ slightly in their effects. PHA stimulates mature T cells, Con A mature and immature T cells, and PWM both T and B cells. After 2.5 months of dietary treatment, high fat had no effect on splenocytes in groups fed SF or PSF, whereas the group fed 24% PUFA had a depressed response to all three mitogens as compared to the group fed the control. The depression in [^3H]thymidine incorporation was statistically significant ($p \leq 0.01$) at the mitogen dose giving peak proliferation, and the same trend was present at two other doses of Con A and PWM. This depressed response due to PUFA was even more highly significant when the transformation was expressed as the stimulation index (SI) rather than as cpm.

A different effect of diet was noted on the MLN. The PSF fed for 2.5

months enhanced T-cell activation by Con A at all doses used. There was, however, much variation, although the effect was marginally statistically significant ($p < 0.07$) when data were log transformed. The MLN proliferation was not markedly affected by dietary fat using PHA or PWM and PUFA had no depressive effect as seen in splenocytes.

When cells were cultured in rat serum there was a higher mitogen requirement to achieve peak transformation. Moreover, the PUFA depressant effect was no longer present when rat serum was used for culture. The authors report that the mitogen-stimulated response was not markedly different in any dietary group when splenocytes were cultured in the presence of rat serum.

In the 5-month study the depressed splenic lymphocyte response due to PUFA persisted but was not statistically significantly different. There were no differences between the control, the SF, and the PSF. Again the MLN transformation differed from the splenic response. It also differed from the response after the 2.5-month feeding trial. Both SF and PSF gave an enhanced response to Con A and PWM but not to PHA. Apparently because of a large variability among animals, however, these findings were not statistically different. The histopathological findings in this study were that high fat diets caused significant splenic hyperplasia but no consistent changes in the mesenteric tissue.

This last study is significant in drawing attention to the fact that the effect of fat on the immune system depends on the duration of feeding, the type of fat used, and the immune organ examined. It also emphasizes that dose responses to the mitogens must be conducted to determine the peak response for each group. It also provides a partial explanation for conflicting reports. Ossman *et al.* (1980) found in mice that if dietary corn oil was increased in the order 8, 14, and 20% the splenic response to Con A but not PWM was enhanced whereas the 20% corn oil reduced the response to lipopolysaccharide (LPS). Eight and 14% beef tallow diets stimulated the Con A response but had no effect with PWM and LPS. Responses to saturated fat were variable, but 6-week-old animals showed higher responses to Con A. A report by Erickson *et al.* (1980) agrees that PUFA compared to SF (coconut oil) depresses blastogenesis induced by Con A. On the other hand, Kollmorgan *et al.* (1979) reported no effect of PUFA on Con A-induced lymphocyte transformation. Differences in length of the feeding trial, mitogen dose, types of fat (i.e., beef tallow versus coconut oil), and choice of tissue could all contribute to differences in results. The Kollmorgan *et al.* study was further complicated by the use of horse serum in the medium for splenocyte culture, which probably abrogated the dietary effects. A report by Friend *et al.* (1980) suggested that high fat diets inhibit lymphocyte responses to mitogens. Un-

fortunately, the authors used a closed-formula ration as a control group and added fats and casein to this ration, thus diluting out minerals and vitamins. The use of commercial stock rations as control diets and the addition of test nutrients to them is not an uncommon finding. It is, however, poor nutritional practice. Commercial rations frequently differ with the prevailing cost of ingredients and may differ from batch to batch. Moreover, such rations are likely to contain various meals from plant and animal sources which may contain materials having an influence on immune responses.

Levy *et al.* (1982) studied the effect of high fat diets on the immune response, production of antiviral factors, and immune complex disease in N2B/N2W (B/W) mice. In this study the diets were extremely well designed. Protein, minerals, vitamins, and fiber were given in the same amounts in diets differing only in the fat. One group received a highly saturated fat in the form of lard, another highly unsaturated (corn oil), and the third low PUFA, also corn oil. Diets II and III differed from diet I in having 10 and 20% more calories, respectively. Mice were sacrificed after 3, 4, and 6 months on the diets. Those mice surviving for 8 months were sacrificed and their kidneys were examined for signs of immune complex disease, which in this strain is characterized by hyperactive B cells producing various autoantibodies. Mice fed the diets high in either SFA or PUFA had more severe immune complex nephritis and an earlier death than mice fed the low-fat diet. However, the serum lipoproteins associated with antiviral activity were increased as high as 1:600,000 in mice on high-fat diets. The high-SFA diet produced higher levels of natural thymotoxic autoantibody, an IgM class of antibody, than did the high-PUFA or low-fat diets. Mice fed high PUFA made more antibodies to double-stranded DNA, an IgG antibody, than did mice on the other two diets.

Levy *et al.* also determined splenocyte blastogenic response to the T cell mitogen Con A. The results were expressed as cpm rather than as the SI. High SFA gave a decreased response compared to high PUFA and low fat at 4 months, but later the response of high PUFA dropped markedly. This change was explained as being due to the very high level of serum lipoproteins associated with antiviral activity. The possibility that the change was due to PG was examined, however. Spleen cell cultures of 4×10^5 cells were treated with $2.8 \mu\text{M}$ indomethacin. This reduced the PGE_2 levels in the culture supernatant fluid by about one half. No effect on Con A response was noted in the SFA group, but in the high PUFA group indomethacin increased the response by 67% in the 3-month samples and by 46% in the 6-month samples. The authors concluded that PGE_2 was not a major factor in the suppression of mitogenesis. Whether this conclusion is sound is difficult to tell because no standard errors of

the means were reported for either the blastogenesis or the PGE₂ levels. Additionally, the indomethacin was insufficient to shut off PGE₂ synthesis. It is also unlikely that a PG effect can be demonstrated using Con A as a mitogen. It has been demonstrated that Con A and PHA may not be sensitive to PG and cAMP suppression (Novogrodsky *et al.*, 1979). When mitogens are used, the PG effect may be bypassed and the change seen presumably reflects changes in the membrane. The finding that high-PUFA diets decrease the mitogenic response is, however, in agreement with Olson *et al.* (1983), who showed that diets high in soybean oil suppress the mitogenic response when compared to the response in a low soybean oil group.

T-cell-mediated responses other than the response to mitogens have been studied. DeWille *et al.* (1981b) compared delayed-type hypersensitivity (DTH) in A/J mice fed an EFA-deficient diet to those fed an EFA-replete diet. After 70 days DTH was reduced by 30% in mice fed the EFA-deficient diet. The reduction of DTH was readily reversed within 7 days when the mice were changed to the control diet. DTH is an *in vivo* measure of T cell cell function based on the inflammatory reaction arising from complex interactions between sensitized T cells, macrophages, and possibly B cells. We also found a decreased DTH in EFA-deficient Lewis rats (23%) using bovine serum albumin as the sensitizing agent and ear thickness changes (Boissonneault, unpublished results, 1982). DeWille *et al.* measured the response to dinitrofluorobenzene and measured ¹²⁵I uptake in the ear.

Several investigators have examined B cell responses in animals fed different dietary fats. The Jerne plaque-forming cell (PFC) assay has been used to assess antibody production by single cells. Erickson *et al.* (1983) found that as the linoleic acid levels in the lymphocyte increased (i.e., higher PUFA diets) the direct (IgG) PFC response to sheep red blood cells (SRBC) decreased. This finding was in contrast to the finding of DeWille *et al.* (1979), who reported that in A/J mice EFA deficiency caused a decreased response and increased energy from dietary fat above 2% did not significantly increase the response. In our laboratory we have examined the PFC response to SRBC in EFA-deficient Lewis rats and, in agreement with Erickson *et al.* found that the deficiency increased the response. It should be noted that both Erickson *et al.* and DeWille *et al.* used an intraperitoneal (ip) injection of SRBC, whereas we employed the intravenous (iv) route. We attempted to resolve the discrepancy between our results and those of DeWille *et al.* by studying two differences in the studies which could have accounted for it; namely, the route of injection of SRBC and the species of animal. The results revealed no differences between species. Both EFA-deficient Lewis rats and A/L mice showed an

increase in PFC response when the antigen was injected iv and no difference between EFA-deficient and EFA-replete animals when an ip route was used (Boissonneault and Johnston, 1984). The route of injection of antigen is known to change the magnitude of immune responses (Morse, 1965). However, Erickson *et al.* found the increased PFC response in their deficient animals using an ip injection route. They explain their results in terms of a change in membrane fluidity due to dietary fat. They state that there is an inverse relationship between tissue linoleic acid content and PFC response. Our studies, however, show that the inverse relationship applies more correctly to tissue arachidonic acid levels and capacity for PG synthesis. We found a direct correlation between a decreased prostaglandin $F_{2\alpha}$ ($PGF_{2\alpha}$) synthesis by spleen and an increased direct PFC response (Boissonneault and Johnston, 1983). The correlation between $PGF_{2\alpha}$ synthesis and PFC response was confirmed by the fact that if EFA-replete animals were given an injection of the PG-synthase inhibitor indomethacin their spleen cells showed a significant increase in PFC response (Table VI). This PG effect does not, however, preclude that there is also an effect on membrane properties.

Our finding of a decreased PG synthesis correlated with an increased humoral response is consistent with an immunosuppressive effect of PG. The mechanism by which reduced PG synthesis leads to increased immune responsiveness to a xenogenic target cell like SRBC has not been defined. If PGs are involved in T suppressor activity as suggested by

Table VI
EFFECT OF ESSENTIAL FATTY ACID DEFICIENCY AND INHIBITION OF CYCLOOXYGENASE ON SPLENIC PROSTAGLANDIN PRODUCTION AND THE DIRECT PLAQUE-FORMING CELL RESPONSE IN LEWIS RATS^{a,b}

Diet	Indomethacin pretreatment	Splenic PGE ^c (n = 4)	Splenic PGF ^c (n = 4)	PFC ^d (n = 5)
EFA-deficient	—	1242 ± 864 ^e	739 ± 9	690 ± 144
EFA-adequate	—	1475 ± 150	1228 ± 159	490 ± 48
EFA-deficient	0.4 mg	1355 ± 370	989 ± 306	648 ± 127
EFA-adequate	1.0 mg	976 ± 160	445 ± 153	1022 ± 270

^a Reprinted with permission from Johnston, P. V., and Marshall, L. A. (1984). Dietary Fat, Prostaglandins and the Immune Response. *In* Progress in Food and Nutrition Science, Vol. 8. Pergamon, Oxford.

^b Abbreviations: essential fatty acid, EFA; plaque-forming cell, PFC; prostaglandin E, PGE; prostaglandin F, PGF.

^c Expressed as pg/mg splenic protein as determined 5 minutes after 5×10^8 SRBC iv.

^d PFC/ 10^6 cells.

^e Mean ± SD.

Webb and Nowowiejski (1978), it may be related to a decrease in T suppressor activity which would lead to an unregulated response (Fulton and Levy, 1980). NZB/W mice have a high incidence of systemic lupus erythematosus; they lose their ability to respond to SRBC with increase in PG with age and lose T suppressor function (Zurier *et al.*, 1977).

In summary, high PUFA has been generally shown to depress the mitogenic response of lymphocytes, but the result is not the same for all sources of cells. Duration of feeding, age of onset of feeding, and the organ examined all influence the results. Mitogenesis is enhanced in EFA deficiency and PFC response has been reported as decreased or increased in deficient rats and mice. Delayed-type hypersensitivity is reduced in EFA deficiency but is readily reversed on refeeding EFA. Both the PFC and the DTH responses may be related to decreased PG production in EFA deficiency. The effect on mitogenesis has largely been explained on the basis of changes in the lipid environment of the plasma membranes. That there are PG effects cannot be ruled out because the mitogens used, such as Con A and PHA, bypass the early events of antigen response and stimulate lymphocytes in such a way that suppression by PG or cAMP is relatively insensitive.

B. DIETARY $n - 6$ AND $n - 3$ FATTY ACIDS AND EICOSANOID SYNTHESIS BY IMMUNOCOMPETENT CELLS

In the last few years the status of $18:3n - 3$ as an essential nutrient has been reevaluated. Except under unusually stressful circumstances such as total parenteral nutrition (TPN) without an $18:3n - 3$ source, clinical signs of an $18:3n - 3$ deficiency are not likely to occur. However, we and others have adopted the view that, although $18:3n - 3$ deficiency does not manifest readily in a clinical form, it may be essential in the sense of having a function which involves modulation of $18:2n - 6$ metabolism and therefore of eicosanoids. This possible function lies in the fact that the Δ^6 desaturase involved in the synthesis of the PUFA precursors of PG prefers the $n - 3$ fatty acid as a substrate before the $n - 6$ and the $n - 9$. Hwang and Carroll (1980) reported a modulatory effect of dietary $18:3n - 3$ on PG synthesis by platelets, and Galli *et al.* (1978) in cerebral cortex. To determine if a function for $18:3n - 3$ may be its ability to modulate the synthesis of PG in systems in which it is regulatory, we selected the immune system as a model.

Various levels of $18:2n - 6$ and $18:3n - 3$ were fed to rats over several time periods. The composition of the diets and the ratios of $18:2n - 6$ to $18:3n - 3$ are shown in Table VII. Male and female rats were fed one of three or four of the diets for periods of 2 months, 4 months, or through the

Table VII
COMPOSITION OF DIETS VARYING IN THE RATIO OF 18:2*n* - 6 TO 18:3*n* - 3

Component	Diet (g/100 g)			
	Corn oil	Soybean oil	Soybean/linseed ^a oil	Linseed oil
Corn oil ^b	10	—	—	—
Soybean oil ^c	—	10	2.3	—
Linseed oil ^d	—	—	7.7	10
Casein	20	20	20	20
Dextrose	18	18	18	18
Cornstarch	30.5	30.5	30.5	30.5
Cellulose	5	5	5	5
Salt mix	4	4	4	4
Vitamin mix	12.5	12.5	12.5	12.5
18:2 <i>n</i> - 6/18:3 <i>n</i> - 3	32:1	7:1	1:1	1:3

^a Fatty acid analysis: 16:0, 1.0%; 18:0, 3.1%; 18:1*n* - 9, 20.0%; 18:2*n* - 6, 38.2%; 18:3*n* - 3, 37.7%.

^b Fatty acid analysis: 16:0, 8.7%; 18:0, 1.4%; 18:1*n* - 9, 16.5%; 18:2*n* - 6, 51.0%; 18:3*n* - 3, 2.2%. Courtesy of Best Foods, CPC International, Inc., Englewood Cliffs, NJ.

^c Fatty acid analysis: 16:0, 11.8%; 18:0, 3.7%; 18:1*n* - 9, 20.3%; 18:2*n* - 6, 56.3%; 18:3*n* - 3, 7.7%. Courtesy of Kraft, Inc., Memphis, TN.

^d Fatty acid analysis: 16:0, 5.5%; 18:0, 2.6%; 18:1*n* - 9, 12.5%; 18:2*n* - 6, 17.5%; 18:3*n* - 3, 61.8%. Courtesy of Cargill, Inc., Minneapolis, MN.

second generation. Fatty acid profiles of ethanolamine phosphoglycerides (EGP) on thymus and spleen were determined. The fatty acid patterns of liver and brain slices were also determined in order to compare the degree of change in immunocompetent tissue with a tissue that responds readily to diet, like the liver, and one resistant to dietary change, the brain. No adverse effects of feeding the higher 18:3*n* - 3 diets were noted on weight gain, feed intake, or reproductive performance.

In 2 months the EGP of liver showed a decrease in 20:4*n* - 6 as the 18:3*n* - 3 content increased relative to 18:2*n* - 6. The 20:5*n* - 3 rose from nondetectable levels on the corn oil diet to 12% on the linseed oil diet. As 18:3*n* - 3 was increased, *n* - 6 fatty acids with the exception of 18:2*n* - 6 decreased and *n* - 3 fatty acids, especially 22:6*n* - 3, increased. The spleen responded somewhat differently in that the 20:5*n* - 3 level did not change appreciably. After a 4-month feeding period there was little change in the magnitude of the fatty acid alterations (Marshall and Johnston, 1982). Some of the results on the effect of the diets on the synthesis of the major PG of the tissue are shown in Table VIII. All

Table VIII
PROSTAGLANDIN SYNTHESIS BY LIVER, SPLEEN, AND THYMUS HOMOGENATES AND
BRAIN SLICES OF RATS FED DIFFERENT RATIOS OF 18:2n - 6 TO 18:3n - 3

	Corn oil	Soybean oil	Soybean-Linseed oil	Linseed oil
	PGF _{2α} (pg/mg tissue) ^a			
Diet ^b				
2 month				
Liver ^c	470.1 ± 56.6 ^d (10)	441.4 ± 32.2 (7)	298.6 ± 28.9 (10)	195.6 ± 20.7 (9)
	PGE ₂ (pg/mg tissue) ^e			
Thymus ^f	122.2 ± 25.8 (9)	103 ± 11.4 (8)	100.8 ± 20.6 (9)	86.6 ± 10.6 (9)
Spleen ^f	83.0 ± 26.6 (4)	—	63.3 ± 8.7 (7)	45.9 ± 11.8 (8)
Second generation				
Spleen ^g	106.5 ± 18.9 (6)	103.4 ± 18.5 (9)	59.3 ± 8.1 (11)	
	ng PGF _{2α} /100 mg wet wt ^e			
Brain slices ^h	3.4 ± 0.7 (4)	2.5 ± 0.4 (4)	1.5 ± 0.2 (10)	

^a Synthesis after 15-min incubation, corrected for zero-time synthesis.

^b Diets containing 10% corn oil (CO), soybean oil (SO), soybean/linseed oil mix (SL), or linseed oil (LO).

^c CO significantly different from SL and LO ($p < 0.01$); SO significantly different than SL and LO ($p < 0.01$).

^d Mean ± SEM.

^e Synthesis after 30 min-incubation, corrected for zero-time synthesis.

^f Not significant.

^g CO and SO significantly different from SL ($p < 0.01$).

^h CO and SO significantly different from SL ($p < 0.01$).

tissues except the brain showed a decrease in PG synthesis after 2 or 4 months on the diets. The brain, which is known to be resistant to diet-induced changes, did not show an effect until the diet had been fed through the second generation.

Whole tissues were used in the above studies. The effects of the diets on the immunocompetent cells are therefore somewhat masked by nonimmune tissue. More recently we have completed studies on isolated cell populations. As noted earlier (Table III) the fatty acid changes seen are much more marked when these cell populations are examined. The effects

on PG synthesis were pronounced when either an EFA-deficient diet or high ratios of $18:3n-3$ to $18:2n-6$ were fed. In both unstimulated and PHA-stimulated peripheral blood lymphocytes and splenocytes the deficient diet depressed PG synthesis and a linseed oil diet depressed it even further (Marshall and Johnston, unpublished results). The cells were cultured with PHA for 48 hours and the PG measured should largely be due to T cells. Synthesis by neutrophils is, however, a possibility. This study does not, therefore, provide proof that immune cells other than macrophages produce PG, which would require a rigidly purified T or B cell preparation. That the elevated levels of $20:5n-3$ and reduced $20:4n-6$ led to a depression in PG synthesis is consistent with the finding that $20:5n-3$ is an effective competitor for the cyclooxygenase (Culp *et al.*, 1979; Lands, 1981). In order to determine that the effect seen was definitely due to the dietary alteration in the phospholipid fatty profile and not to an increased catabolism of PG or to a membrane effect unrelated to the dietary changes, the following studies were conducted. The cyclooxygenase inhibitor indomethacin, together with PGE_2 or $20:4n-6$, was added to cultures. The addition of indomethacin and PGE_2 led to a marked depression of PG synthesis in all groups, and there was no difference between groups. Thus, the changes seen were due to a difference in synthesizing capacity and not to increased breakdown of PGE_2 . The addition of the $20:4n-6$ abrogated the dietary effects. The decreased PG-synthesizing capacity of cells from groups receiving higher levels of $18:3n-3$ is therefore due to the competition of $18:3n-3$ with $18:2n-6$ for the Δ^6 desaturase and competition of its metabolite, $20:5n-3$, with the $18:2n-6$ metabolite $20:4n-6$.

Because the macrophage is a major source of PG and other eicosanoids, we have conducted a number of studies using rat peritoneal macrophages. Male, weanling Sprague-Dawley rats were fed for 6 weeks either the corn oil or the linseed oil diet previously described (Table VII). The peritoneal cells were collected by injecting anesthetized rats with phosphate-buffered saline. The abdomen was massaged and the peritoneal cells withdrawn with a syringe. By adherence techniques a macrophage population which was greater than 90% pure, as judged by morphology and α -naphthyl acetate esterase staining, was obtained.

As noted in the other cells, there was an appreciable increase in $n-3$ fatty acids in the macrophages from the linseed oil group and a decrease in $n-6$ fatty acids. The linseed oil and corn oil group macrophage EGP and choline glycerophosphatide (CGP) contained 7.4 and 5.5% $20:5n-3$ and 19.3 and 13.9% $20:4n-6$, respectively, compared to 0.1 and 0.2% $20:5n-3$ and 31.5 and 28% $20:4n-6$, respectively. The level of $20:5n-3$ in serine glycerophosphatide increased from 0.6 to only 1.4%, and there was

less change in 20:4n - 6 than in EGP and CGP. The fatty acid changes were associated with a decreased PGE₂ synthesis. Synthesis was determined at 1 hour and 4 hours after stimulation with zymosan and culture in medium containing 5% homologous rat serum. The PGE₂ synthesis is shown in Table IX. Studies with added inhibitor and PGE₂ or added 20:4n - 6 or 20:5n - 3 again showed that there was no increased catabolism of PGE₂ and the addition of the competing precursors abrogated dietary effects.

Some tests of function were included in the macrophage study. The changes induced by feeding increased levels of 18:3n - 3 did not affect phagocytosis of either yeast cells (*Saccharomyces cerevisiae*, magna) or carbon particles. These methods give a percentage of macrophages phagocytosing the particles and do not preclude that there is a PG effect at some stage of the phagocytic process. The decrease in PG production by the linseed oil macrophages did correlate with less arginase activity when compared to the corn oil macrophages. Arginase activity is thought to be part of a nonspecific cytotoxicity shown by activated macrophages (Kung *et al.*, 1977; Allison, 1978).

An interesting report on the effect of Intralipid (Cutter) on macrophages was published by Strunk *et al.* (1983). Intralipid, the commonly used fat source in total parenteral nutrition (TPN), is an emulsion of soybean oil. Intralipid is known to be taken up by macrophages, and Strunk and co-workers studied this effect on the macrophage membrane fatty acid composition. The macrophage is the site of synthesis of several proteins of the

Table IX
SYNTHESIS OF PROSTAGLANDIN E₂ BY
PERITONEAL MACROPHAGES FROM RATS FED
DIFFERENT RATIOS OF 18:2n - 6 TO 18:3n - 3^a

Culture times (hours)	Corn oil	Linseed oil
1	1451 ± 107 ^b (6)	455 ± 61 ^c (8)
4	3554 ± 431 (6)	1020 ± 160 ^c (8)

^a PGE₂ in pg/ml culture medium. Approximately 1 × 10⁶ adherent macrophages stimulated with zymosan in a final volume of 2 ml of culture medium. Number of animals indicated in parentheses.

^b Mean ± SEM.

^c Linseed oil significantly different from corn oil (*p* > 0.0001).

complement system, so they also examined the effect of Intralipid on this system in the guinea pig. Macrophage membrane phospholipids were found to have increased levels of palmitic, oleic, and linoleic acids and decreased levels of $20:4n-6$ after incubation with Intralipid. Macrophages incubated in Intralipid (5–40 mg/dl) produced less C_2 and C_4 compared to controls. Human macrophages incubated in 19 or 38 mg/dl showed decreases of 45 and 50%, respectively, of C_2 . This effect could be reversed by incubating in the presence of both Intralipid and $20:4n-6$. The authors believe that this is not a PG-related effect because PGE_2 synthesis was not altered by incubation with Intralipid. The reversal was not found if the cells were incubated with $20:4n-6$ alone. A PG analysis does not appear to have been done on the cells cultured in both the emulsion and $20:4n-6$. It would have been of interest to see if PG synthesis was, in this case, stimulated. The report has implications in TPN. The authors suggest that it would be wise to keep Intralipid concentrations to less than 20 mg/dl in infants or, alternately, to use intermittent infusions to give time for macrophages to clear the lipid, a practice which is not uncommon in TPN. Intralipid has been shown to cause fatty liver due to the uptake by the macrophages.

Most studies on dietary fat and immunity have considered only the stable PG, and there have been few studies on the effects of lipoxygenase products. For two reasons this is now likely to change rapidly. Firstly, it is appreciated that the entire eicosanoid profile must be examined if a dietary precursor is altered in the diet. Secondly, the technical difficulties involved in examining the whole profile are being overcome as high-performance liquid chromatographic procedures are being developed (Terragno *et al.*, 1981; Morris *et al.*, 1983; Peters and Schulman, 1983), and antisera are becoming available to some of the lipoxygenase products. It is possible that, if $20:4n-6$ is diverted from the cyclooxygenase pathway by a competing nutrient or by an enzyme inhibitor, events may be altered in the lipoxygenase pathway. This diversion may not necessarily take place but it has, as yet, been insufficiently studied to judge its significance. It is, however, a very important consideration because dietary manipulations which may bring about apparently desirable effects on one function may have deleterious effects on another.

One study has addressed the question of the effect of dietary fat on the production of tetraene and pentaene leukotrienes (Murphy *et al.*, 1981). A mastocytoma cell line was grown in mice that had been raised on a diet rich in $20:5n-3$ provided by menhaden oil. Whole harvested cells were stimulated with calcium ionophore A23187 to produce lipoxygenase products. Leukotrienes B_4 , B_5 , C_4 , and C_5 were isolated and characterized by high-performance liquid chromatography (HPLC), UV absorption, and

mass spectrometry. The $20:4n - 6$ in the mast cell tumor lipids fell from 9.2 to 3.9 mol% and $20:5n - 3$ increased from 0.5 to 4.5 mol% when the mice had been fed the menhaden oil. The relative amounts of leukotrienes B_4 and B_5 (LTB_4 , LTB_5) were associated with the relative amounts of the precursors, suggesting not only that the epoxide leukotriene (LTA) can be derived from either $20:4n - 6$ or $20:5n - 3$ but also that dietary fat can influence leukotriene production. The dietary design of this study was not entirely satisfactory in that the fish oil diet was a fat-free rodent diet with an added 5% menhaden oil and the control used was a commercial ration. The unsatisfactory nature of this type of control has previously been noted. However, the study does illustrate two important points. It shows that the synthesis of lipoxygenase products from dietary $n - 3$ fatty acids may be of significance and it emphasizes the need to study entire eicosanoid profiles.

There have been few studies of dietary fat and immunity in humans. Utermohlen and co-workers have fed safflower oil or olive oil to multiple sclerosis patients and to healthy control subjects. For 5 weeks the subjects were given 21 g/day of the oils. Safflower contains over 70% $18:2n - 6$ and less than 1.0% $18:3n - 3$, whereas olive oil contains as much as 70% oleic acid ($18:1n - 9$), about 7–10% $18:2n - 6$, and no $18:3n - 3$. They measured CMI using the migration inhibition assay, which is a measure of lymphokine activity specifically involving the migration inhibition factor. Migration was inhibited with lymphocytes from the olive oil group and less so with lymphocytes from the safflower oil. Agglutination with measles antigen and Con A was increased in the safflower oil but not in the olive oil group (Utermohlen *et al.*, 1982).

IV. Conclusions

There is considerable evidence that eicosanoids are involved in modulation of the immune response, but no clear mechanisms have emerged. A number of complexities and technical difficulties have contributed to the somewhat confusing picture that we presently have in the area. The immune system itself is complex and consists of several cell types with various subtypes. Frequently, studies are conducted with cell populations that are not pure, and the results obtained are the overall effect of interactions between different cell populations. A clear picture can be obtained only when some major questions have been answered. For example, do both macrophages and lymphocytes produce eicosanoids or is the macrophage the sole producer and the effects seen the result of macrophage–lymphocyte interactions mediated via macrophage PG? Another problem

is that studies indicating that PGs have a role in immunoregulation have frequently employed exogenous addition of PG in the range 10^{-5} – 10^{-4} M, which is well outside the range of 10^{-9} – 10^{-8} M which can be considered physiological. On the other hand, dietarily modified capacity for PG synthesis has been correlated with altered immune function, but in only a few instances has the correlation been convincingly demonstrated. Many of the functional changes observed, such as in lymphocyte transformation, may be equally well explained on the basis of change in plasma membrane lipid environments, with consequent changes in enzyme activities, local microviscosity, and receptor properties.

Until recently, technical difficulties have limited the extent to which investigators could examine the wide array of molecules which can result from the oxygenation of eicosanoid precursors. It has become increasingly obvious that there is a need to examine the entire eicosanoid profile. These powerful biological molecules can have opposing effects, and a molecule produced in one pathway may have an inhibitory effect on the production of another. There is a particular need to know to what extent the trienoic eicosanoids are produced on diets differing in the level of $18:3n-3$ to $18:2n-6$. When dietary $n-3$ fatty acids are high it now appears that at least some trienoic PG is produced, but the extent to which other eicosanoids are formed is unknown. Even given the recent improvement in analysis of eicosanoids, we need to recognize that there are limitations in our ability to determine minute quantities. There is need for improvement in both sensitivity and specificity of methods.

Well-designed dietary studies could help elucidate the role of eicosanoids in immunoregulation. The capacity for eicosanoid synthesis can be modified by dietary means and the changes can be brought about in a relatively short time (Marshall *et al.*, 1983). Both *in vivo* and *in vitro* tests of immune function can be conducted and the response of animals to challenge with infectious agents can be studied. Many of the studies conducted thus far have been flawed in dietary design and in execution and interpretation of assays. Major errors in assays have included culture of cells in serum other than homologous or autologous, with resultant fatty acyl exchanges and abrogation of dietary treatments. As noted by some investigators, it is essential that assays involving mitogens or antigens should include a dose response curve for each dietary group, because assays at a single dose can give misleading results. It also must be kept in mind that cells from different sources can respond differently to the dietary variable. Peripheral lymphocyte populations and lymph node cells can be expected to respond to changes in dietary fat more rapidly than spleen cells.

This article has mainly been concerned with studies on the effect of

dietary fat, both its quantity and quality, on eicosanoid synthesis and immune function. A number of recommendations for improvement of dietary protocol can be made. Commercial rations are not appropriate controls for studies on dietary fat variables in semipurified diets. The addition of the fat of interest to a commercial feed dilutes all the other nutrients. Diets being compared should differ only in fat content or the fatty acid profiles of the fat. They should be adjusted to be identical in protein, minerals, and vitamins. Fatty acid profiles of each batch of fat or oil should be determined. Periodic checks of the peroxide value of high-PUFA oils should be made. It should also be remembered that high-PUFA diets may require that the vitamin E content be increased above normally recommended levels. When comparing results from one laboratory to another, all differences in the diets should be considered. For example, many minerals are important in immune responses. Zinc is particularly important in dietary fat studies because of the relationship between zinc and essential fatty acids. High dietary iron levels may influence oxygenation events and eicosanoid synthesis. Different types of protein are known to influence immune responses. It is particularly significant that mice fed diets based on casein, soy, and wheat proteins showed substantially lower humoral response than that of mice fed a stock diet, whereas mice fed lactalbumin had a response about five times higher than mice fed casein (Bounos *et al.*, 1983). This finding illustrates how misleading the use of a stock diet as a control can be in immunological studies. The age of onset of the dietary treatments is also known to influence both CMI and humoral immunity, and it appears that differences in the length of a deficiency rather than its severity may explain conflicting reports on effects on immune responses (Wade *et al.*, 1983).

There is considerable evidence that eicosanoids have a modulatory role in immunity, but precisely which cells produce PG and related products is still a matter of controversy. Moreover, at what point in the immune response PGs act is unclear. Lipoxygenase products including LTB_4 have been shown to be involved in chemotactic activity for neutrophils (Ford-Hutchinson *et al.*, 1980) and both LTB_4 and leukotriene D_4 (LTD_4) show some inhibitory effect on the human lymphocyte proliferative response to Con A. Bailey *et al.* (1982) reported similar findings for 15-HETE in mice.

There is also evidence that dietary fat can influence immune responses, but that this is entirely due to altered eicosanoid synthesis is unlikely, although they may be involved at some point and in some cells. Other effects of dietary fat *in vivo* must be considered, such as altered serum lipoproteins (Gurr, 1983) plasma membrane structure and local microviscosity, and hormonal status. At this stage of knowledge no conclusions can be drawn regarding what the relationship between dietary fat and

immunity means in terms of understanding immune-related disorders or their control. Well-designed nutritional models should, however, help elucidate the apparent modulatory roles of eicosanoids, lipoproteins, and other fat-related metabolites in immunity.

ACKNOWLEDGMENTS

Work conducted by the author and associates was supported by a grant from the Science and Education Administration of the US Department of Agriculture Competitive Grants Office No. 59-2171-1-1-633-0 and by work conducted under Project No. ILLU-50-0350 of the Agricultural Experiment Station, College of Agriculture, University of Illinois at Urbana-Champaign.

References

- Aaes-Jørgensen, E. (1982). *Prog. Lipid Res.* **20**, 123-128.
- Allison, A. C. (1978). *Immunol. Rev.* **40**, 3-27.
- Bailey, J. M., Bryant, R. W., Low, C. E., Pupilla, M. B., and Vanderhoeck, J. Y. (1982). *Cell. Immunol.* **67**, 112-120.
- Boissonneault, G. A., and Johnston, P. V. (1983). *J. Nutr.* **113**, 1187-1194.
- Boissonneault, G. A., and Johnston, P. V. (1984). *J. Nutr.* **114**, 89-94.
- Boukhchace, D., and Lagarde, M. (1982). *Biochim. Biophys. Acta* **713**, 386-392.
- Bounos, G., Létourneau, L., and Kongshavn, P. A. L. (1983). *J. Nutr.* **113**, 1415-1421.
- Carroll, K. K., and Khor, H. T. (1975). *Biochem. Pharmacol.* **10**, 308-358.
- Chandra, R. K., and Newberne, P. M. (1977). "Nutrition, Immunity and Infection." Plenum, New York.
- Clausen, J., and Møller, J. (1969). *Int. Arch. Allergy Appl. Immunol.* **36**, (Suppl.), 224-233.
- Clifford, C. K., Smith, L. M., Erickson, K. L., Hamblin, C. L., Creveling, R. K., and Clifford, A. J. (1983). *J. Nutr.* **113**, 669-679.
- Culp, B. R., Titus, B. G., and Lands, W. E. M. (1979). *Prostaglandins Med.* **3**, 269-279.
- DeWille, J. W., Fraker, P. J., and Romsos, D. R. (1979). *J. Nutr.* **109**, 1018-1027.
- DeWille, J. W., Fraker, P. J., and Romsos, D. R. (1981a). *Fed. Proc., Fed. Am. Soc. Exp. Biol.* **40** (Abstr.), 344.
- DeWille, J. W., Fraker, P. J., and Romsos, D. R. (1981b). *J. Nutr.* **111**, 2039-2041.
- Dyerberg, J., and Bang, H. O. (1979). *Lancet* **II**, 433-435.
- Dyerberg, J., Bang, H. O., and Hjørne, N. (1975). *Am. J. Clin. Nutr.* **28**, 958-966.
- Erickson, K. L., Gershwin, M. E., Canolty, N. L., and Eckels, D. D. (1979). *J. Nutr.* **109**, 353-359.
- Erickson, K. L., McNeill, C. J., Gershwin, M. E., and Ossman, J. B. (1980). *J. Nutr.* **110**, 1555-1572.
- Erickson, K. L., Adams, D. A., and McNeill, C. A. (1983). *Lipids* **18**, 468-474.
- Falardeau, P., Hamberg, M., and Samuelsson, B. (1976). *Biochim. Biophys. Acta* **441**, 193-200.
- Falardeau, P., Oates, J. A., and Brash, A. R. (1981). *Anal. Biochem.* **115**, 359-367.
- Ferber, E., and Resch, K. (1977). In "The Lymphocyte: Structure and Function" (J. J. Marchalonis, ed.), pp. 593-621. Dekker, New York.
- Ferber, E., Kröner, B., Fischer, H., Peskar, B. A., and Anders, C. (1980). In "Membrane

- Fluidity, Biophysical Techniques and Cellular Recognition" (M. Kates and A. Kuksis, eds.), pp. 239–263. Humana Press, Clifton, NJ.
- Fischer, S., and Weber, P. C. (1984). *Nature (London)* **307**, 165–168.
- Ford-Hutchinson, A. W., Bray, M. A., Doig, M. V., Shipley, M. E., and Smith, M. J. H. (1980). *Nature (London)* **286**, 264–265.
- Friend, J. V., Lock, S. O., Gurr, M. I., and Parish, W. E. (1980). *Int. Arch. Allergy Appl. Immunol.* **62**, 292–301.
- Fulton, A. M., and Levy, J. G. (1980). *Cell. Immunol.* **52**, 29–37.
- Galli, G., Spagnuolo, C., Bosio, E., Tosi, L., Folco, G. C., and Galli, C. (1978). *Adv. Prostaglandin Thromboxane Res.* **4**, 181–189.
- Goldyne, M. E., and Stobo, J. D. (1981). *CRC Crit. Rev. Immunol.* pp. 189–223.
- Goodwin, J. S., and Webb, D. R. (1980). *Clin. Immunol. Immunopathol.* **15**, 106–122.
- Goodwin, J. S., Bankhurst, A. D., and Messner, R. P. (1977). *J. Exp. Med.* **146**, 1719–1734.
- Guarnieri, M., and Johnson, R. M. (1970). *Adv. Lipid Res.* **8**, 115–174.
- Gurr, M. I. (1983). *Prog. Lipid Res.* **22**, 257–287.
- Habu, S., and Raff, M. L. (1977). *Eur. J. Immunol.* **7**, 451–457.
- Holman, R. T., Johnson, S. B., and Hatch, T. F. (1982). *Am. J. Clin. Nutr.* **35**, 617–623.
- Hwang, D. H., and Carroll, A. E. (1980). *Am. J. Clin. Nutr.* **33**, 590–597.
- Kennerly, D. A., Sullivan, T. J., Sylwester, P., and Parker, C. W. (1979). *J. Exp. Med.* **150**, 1039–1044.
- King, M. A., and Spector, A. A. (1981). In "The Growth Requirements of Vertebrate Cells *in vitro*" (C. Weymouth, R. Harm, and P. J. Chapple, eds.), pp. 293–312. Cambridge Univ. Press, London and New York.
- Kollmorgen, G. M., Sansing, W. A., Lehman, A. A., Fischer, G., Longley, R. E., Alexander, S. S., King, M. M., and McCay, P. B. (1979). *Cancer Res.* **39**, 3458–3467.
- Kung, J. T., Brooks, S. B., Jakway, J. P., Leonard, L. L., and Talmadge, D. W. (1977). *J. Exp. Med.* **146**, 665–672.
- Lagarde, M., Gharib, A., and Dechavanne, M. (1977). *Biochimie* **59**, 935–937.
- Lampert, M. S., and Walker, B. L. (1976). *J. Nutr.* **106**, 86–93.
- Lands, W. E. M. (1981). In "Nutritional Factors: Modulating Effects on Metabolic Processes" (R. F. Beers and E. Bassett eds.), pp. 489–493. Raven, New York.
- Levy, J. A., Ibrahim, A. B., Shirai, T., Ohta, K., Nagasawa, R., Yoshida, H., Estes, J., and Gardner, M. (1982). *Proc. Natl. Acad. Sci. U.S.A.* **79**, 1974–1978.
- Lockniskar, M., Nauss, K. M., and Newberne, P. M. (1983). *J. Nutr.* **113**, 951–961.
- Loomis, R. J., Marshall, L. A., and Johnston, P. V. (1983). *J. Nutr.* **113**, 1292–1298.
- Maino, V. C., Hayman, M. J., and Crumpton, M. J. (1975). *Biochem. J.* **146**, 247–252.
- Marshall, L. A., and Johnston, P. V. (1982). *Lipids* **17**, 905–913.
- Marshall, L. A., and Johnston, P. V. (1983). *Lipids* **18**, 737–742.
- Marshall, L. A., Szczesniowski, A., and Johnston, P. V. (1983). *Am. J. Clin. Nutr.* **38**, 895–900.
- Mathias, M. M., and Dupont, J. (1979). *Lipids* **14**, 241–252.
- Meade, G. J., and Mertin, J. (1978). *Adv. Lipid Res.* **16**, 127–165.
- Meerphol, H.-G., Lohmann-Matthes, M. L., and Fischer, H. (1976). *Eur. J. Immunol.* **6**, 213–217.
- Morris, H. R., Clinton, P. M., Taylor, G. W., Piper, P. J., Tippins, J. R., and Barnett, K. (1983). In "Leukotrienes and Other Lipooxygenase Products" (P. J. Piper, ed.). Wiley, New York.
- Morse, S. I. (1965). *J. Exp. Med.* **121**, 49–68.
- Munder, P. G., Ferber, E., Modellel, M., and Fischer, H. (1969). *Int. Arch. Allergy Appl. Immunol.* **36**, 117–128.

- Murphy, R. C., Pickett, W. C., Culp, B. R., and Lands, W. E. M. (1981). *Prostaglandins* **22**, 613-622.
- Nacchace, P. H., Sha'afi, R. I., Borgeat, P., and Goetzl, E. J. (1981). *J. Clin. Invest.* **67**, 1584-1587.
- Needleman, P., Minkes, M., and Raz, A. (1976). *Science* **193**, 163-165.
- Needleman, P., Raz, A., Minkes, M. S., Ferrendelli, J. A., and Sprecher, H. (1979). *Proc. Natl. Acad. Sci. U.S.A.* **76**, 944-948.
- Novogrodsky, A., Rubin, A. L., and Stenzel, K. H. (1979). *J. Immunol.* **122**, 1-7.
- Olson, L. M., Clinton, S. K., and Visek, W. J. (1983). *Fed. Proc., Fed. Am. Soc. Exp. Biol.* **42**, 1186 (Abst. 5240).
- Ossman, J. B., Erickson, K. L., and Canolty, N. L. (1980). *Nutr. Rep. Int.* **22**, 279-284.
- Parker, C. W. (1981). In "Mechanisms of Lymphocyte Activation" (K. Resch and A. Kirchner, eds.), pp. 47-57. Elsevier, Amsterdam.
- Peters, S. P., and Schulman, E. S. (1983). *J. Immunol. Methods* **64**, 335-339.
- Ring, J., Seifert, J., Mertin, J., and Brendel, N. (1974). *Lancet* **II**, 1331.
- Schreier, S., Polnasek, C. F., and Smith, I. C. P. (1978). *Biochim. Biophys. Acta* **515**, 375-436.
- Selivonchick, D. P., and Johnston, P. V. (1975). *J. Nutr.* **105**, 288-300.
- Stinnett, J. D. (1983). "Nutrition and the Immune Response." CRC Press, Boca Raton, FL.
- Strunk, R. C., Kunke, K. S., Kolski, G. B., and Revsin, B. K. (1983). *Lipids* **18**, 493-500.
- Taniguchi, N., Miyawaki, T., Moriya, N., Nagaoki, T., Kato, E., and Okuda, N. (1977). *J. Immunol.* **118**, 193-201.
- Terragno, A., Rudzik, R., and Terragno, N. A. (1981). *Prostaglandins* **21**, 101-112.
- Tinoco, J., Babcock, R., Hincenbergs, J., Nedwadowski, I., and Miljanich, P. (1978). *Lipids* **13**, 6-17.
- Traill, K. N., and Wick, G. (1984). *Immunol. Today* **5**, 70-76.
- Utermohlen, V., Coniglio, J., Mao, D., Sierra, J., Smith, R., Besner, G., Hutchins, S., Spitzer, K., Tomasso, J., and Boyar, A. (1982). *Prog. Lipid Res.* **20**, 739-741.
- Wade, S., Lemonnier, D., Bleiberg, F., and Delorme, J. (1983). *J. Nutr.* **113**, 1131-1139.
- Walker, B. L. (1981). *Lipids* **16**, 468-474.
- Webb, D. R., and Nowowiejski, I. (1978). *Cell. Immunol.* **41**, 72-85.
- Zurier, R. B., Danjanou, I., Sayadoff, D. M., and Rothfield, N. F. (1977). *Arthritis Rheum.* **20**, 723-728.